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Data-driven Arsenic contamination detection

*Project report for the course 392039 "Machine Learning
for Water Distribution Networks"*

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1 Introduction

Arsenic is a chemical that can occur in drinking water, where it is usually of geogenic origin¹ [1]. Detecting arsenic contamination in drinking water, including water distribution networks, is essential due to the severe health risks associated with exposure. Arsenic contamination can cause nausea, gastrointestinal disorders, skin lesions, cardiovascular diseases, and several types of cancer, among other health problems [1].

While chemical analysis is a common method for arsenic detection [2], this work investigates a machine-learning-based alternative. We implement and compare an Artificial Neural Network (ANN) and a linear regression model on the **Net1** water distribution network provided by the EPyT-Flow Python package [3]. The ANN is used to model nonlinear relationships between sensor measurements and contamination events, whereas linear regression serves as a baseline.

The main contribution of this work is an integrated pipeline that combines ANN-based arsenic contamination detection with GRASP-based sensor optimization. The ANN identifies contamination events from water quality measurements, while GRASP reduces the number of monitoring locations required to maintain reliable performance. In doing so, the proposed framework addresses both contamination detection and monitoring network optimization.

The results show that the ANN substantially outperforms the linear regression baseline, whose performance was insufficient for reliable contamination detection. The best ANN model achieved a BA of 70.46%, a false positive rate of 22.56%, and a detection latency of **X** for arsenic contamination events. Furthermore, chlorine concentration prediction was highly accurate. The GRASP sensor selection procedure reduced the number of required sensors with only a minor decrease in predictive performance, demonstrating the potential of the proposed ANN-GRASP pipeline for efficient and cost-effective water quality monitoring.

¹Geogenic origin refers to arsenic that is naturally released into groundwater through geological processes such as the weathering and dissolution of arsenic-bearing rocks, sediments, and minerals.

2 Related Works

Arsenic Detection. There are various methods to detect arsenic in drinking water. One common approach is spectroscopic analysis, which exploits the interaction between arsenic compounds and electromagnetic radiation to identify and quantify contaminants [2]. Other methods are based on chemical analysis, where the presence of arsenic is inferred through chemical reactions with specific reagents. Arsenic is known to react with chlorine in water [4]. Therefore, chlorine levels in a water distribution system can be influenced by the presence of arsenic. An abrupt decrease in chlorine concentration may indicate an abnormal consumption of chlorine caused by a contamination event. For this reason, variations in chlorine measurements can be exploited to detect anomalies such as arsenic contamination [5]. In this work, we adopt this indirect detection principle by monitoring chlorine levels through chlorine and flow sensors placed at each node of the simplified **Net1** network, originally distributed as a standard benchmark example with EPANET [6].

Water Event Detection. Classical event detection techniques are widely applied to water distribution systems to identify water-quality-related events and support the delivery of safe drinking water. Early detection is essential both to promptly alert operators and consumers and to enable mitigation actions that limit the spread and impact of contamination events. A common approach consists of computing an anomaly score from sensor measurements and comparing it against a predefined threshold. Measurements whose anomaly score exceeds the threshold are classified as events [5]. One of the simplest methods applied to water distribution systems is linear prediction [7]. This method is computationally efficient and easy to implement, but it struggles to capture the nonlinear dynamics and uncertainties that characterize real water networks. More advanced approaches employ Artificial Neural Networks (ANNs) for event detection [8]. ANNs can learn complex relationships from historical sensor data and are generally more robust to nonlinear behaviour. This idea was further improved by Arad et al. in 2013 through the introduction of a dynamic thresholding mechanism, which adapts to changing network conditions and reduces the number of false alarms. In this work, we follow the anomaly detection paradigm and identify contamination events through deviations in chlorine measurements.

Sensor Placement Optimisation. Our toy network contains only nine nodes.

However, real water distribution systems can contain hundreds or thousands of nodes. For economic and practical reasons, it is infeasible to install a chlorine sensor at every node. Furthermore, some locations may not be suitable for sensor deployment due to space limitations, lack of electrical power, or absence of communication infrastructure [9]. Sensor placement can therefore be formulated as an optimization problem. One common approach is Mixed Integer Programming (MIP), which can produce optimal solutions but often requires long computation times and advanced solvers such as CPLEX or Gurobi. Consequently, MIP may become impractical for large-scale networks. In addition, studies have shown that reliable water quality prediction can often be achieved using sensors at only a subset of nodes, with approximately one third of the network being sufficient in many cases [10]. An alternative approach is the Greedy Randomized Adaptive Search Procedure (GRASP) [11]. GRASP is a metaheuristic that combines randomized construction and local search phases to efficiently explore the solution space. Compared with MIP, it requires significantly lower computational effort, scales better to large networks, and does not depend on commercial optimization solvers. Although GRASP does not guarantee global optimality, we adopted this method as studies have shown that a large proportion of its solutions are either optimal or very close to the optimum.

3 Methodology

The methodology consisted of five stages. First, a constant chlorine injection pattern was applied at node 10. Keeping the injection pattern constant provides a stable baseline, so that contamination-related changes in chlorine concentration can be distinguished from changes caused by varying injection conditions. Second, 20 simulations of the **Net1** water distribution network were performed to create the dataset. Chlorine, flow, and arsenic measurements were saved in NumPy and Excel formats. The arsenic measurements were used only to label contamination events during training and evaluation; they were not used as detector inputs.

Third, two data-driven detection approaches were implemented: a linear predictor and an artificial neural network (ANN). Both approaches use chlorine measurements as surrogate indicators of arsenic contamination. Performance was evaluated using detection accuracy, false-alarm behaviour, and detection latency. Finally, GRASP was used to identify three chlorine-sensor locations that maximize the relative number of correctly detected contamination events.

3.1 Linear predictor

The linear predictor, also called linear regression, was chosen according to McKenna et al. [7]. The method consists of the following steps:

1. The regression model is used to predict the chlorine value for the next point in time, based on previous chlorine and flow values.
2. These predicted values are then compared to the actual chlorine values for that point in time. Formula: difference = predicted value - actual value (adapted from [7]).
3. If there is a high difference between the predicted and the actual value, this point in time is marked as an anomaly, which means that an arsenic contamination likely happened [5]. However, to be able to define a high difference, an appropriate threshold has to be defined first.
4. Find the chlorine sensor positions that make the best predictions for the arsenic contamination using GRASP according to Resende & Silva [11]. According to Diao et al. [10], about 1/3 of the nodes need to have sensors to ensure reliable predictions. Since there are 9 nodes in our model, I chose to find the best three sensors.

It is important to know that the regression has to be trained on data with no arsenic contamination. Otherwise, it will learn to predict chlorine values for an arsenic contamination, which would make finding anomalies impossible.

Finding a threshold, as mentioned above, can be done using the regression model on a dataset without any arsenic contamination and determining the threshold with which the model stops predicting any anomalies. The appropriate threshold according to the algorithm was 3.90. However, this means that for the normal data set, all instances would not be predicted to be an anomaly, producing many false negatives. This is not in the interest of the task, especially since in an arsenic contamination scenario, false negatives could be lethal. Nonetheless, instead of changing the threshold, the exact implementation of the linear predictor task was experimented with. This and the results of these experiments are elaborated in the experiments section.

For GRASP to determine the best chlorine sensor placement, a greedy algorithm, optionally followed by a repair procedure, is first applied. This is followed by a local

search phase and finally a path relinking procedure [11]. In this work, the semi-greedy algorithm proposed by Resende and Silva [11] was used for the construction phase. A repair procedure was applied whenever the greedy algorithm selected more than three sensors. Although this situation was not expected to occur frequently, the repair mechanism was implemented to ensure feasibility.

The local search phase explores the neighborhood of the selected sensor locations to identify improved solutions. High-quality solutions are stored in an elite pool containing up to ten solutions. During path relinking, the solution obtained from the local search is combined with a randomly selected solution from the elite pool. The nodes of the two solutions are gradually made identical, generating intermediate solutions that are evaluated at each step.

If an intermediate solution performs better than the worst solution currently stored in the elite pool, or if the elite pool contains fewer than ten solutions, it is added to the pool together with its objective value. The objective value used throughout the optimization process was the relative number of true positives, as correctly identifying contamination events is the primary objective in an arsenic contamination scenario. Consequently, solutions with higher true positive rates were considered superior. After completion of the search process, the best solution contained in the elite pool was selected as the final sensor placement. The entire GRASP procedure was repeated 50 times.

3.2 ANN

The ANN-based detector was developed to capture nonlinear relationships between hydraulic conditions and chlorine concentrations. For each monitored chlorine node, an ANN predicts the current chlorine concentration based on the current flow measurements, the previous 48 chlorine measurements (corresponding to one day), and cyclical time-of-day features. The difference between the predicted and measured chlorine concentration, referred to as the residual,

$$r(t) = \hat{x}(t) - x(t),$$

is used to identify unusual behaviour in the network. Large residuals may indicate the presence of a contamination event.

To distinguish normal variations from anomalies, dynamic upper and lower thresholds are calculated from a sliding window of recent residuals. For each window,

the mean residual μ and standard deviation σ are computed, and the thresholds are defined as

$$T^+(t) = \mu(t) + PDT \cdot \sigma(t), \quad T^-(t) = \mu(t) - NDT \cdot \sigma(t).$$

An observation is considered anomalous if its residual exceeds these thresholds. To prevent contamination events from immediately affecting the threshold calculation, extreme residuals are excluded from the window statistics using positive and negative filters (PF and NF). These filters define the maximum positive and negative residuals that are retained when computing the threshold statistics.

Five parameters determine the behaviour of the detector: the window size, the positive and negative detection thresholds (PDT and NDT), and the positive and negative filters (PF and NF). These parameters were optimized offline using a genetic algorithm and validation data. Candidate parameter sets were evaluated according to their BA, and the best-performing configuration was selected for the final detector.

This approach follows the dynamic-threshold framework proposed by Arad et al. [12]. However, the underlying prediction models differ substantially. Arad et al. employ six water-quality indicators (total chlorine, electrical conductivity, pH, temperature, total organic carbon, and turbidity) and construct a dedicated ANN for each indicator. In contrast, our framework relies on a single water-quality indicator, namely chlorine concentration.

Rather than building separate models for multiple quality parameters, we train one chlorine-prediction ANN for each monitored chlorine sensor. Each ANN receives the recent chlorine history of its node, hydraulic flow measurements, and time-of-day information as inputs and predicts the current chlorine concentration. Flow measurements therefore act only as predictive features and are not modeled by a separate ANN.

Another important difference lies in the event-detection stage. Arad et al. [12] combine the outputs of multiple water-quality indicators using Bayesian probability updates and alarm fusion. In contrast, the present work focuses on arsenic detection using chlorine-based information only. The residual-based dynamic-threshold detector provides diagnostic information, while a supervised ANN classifier combines chlorine measurements, flow measurements, chlorine trends, and temporal features to estimate the probability of an arsenic contamination event.

In addition to the residual-based detector, a supervised ANN classifier was implemented. The classifier combines information about the current state of the network

(chlorine concentrations and flow measurements), recent trends in chlorine concentrations, and the time of day to estimate the probability of an arsenic contamination event. The final alarm threshold was determined using validation data. Consequently, the residual-based detector provides an interpretable indication of unusual network behaviour, while the classifier produces the final contamination prediction. As with the linear predictor, training, parameter tuning, and final evaluation were performed on separate data sets to avoid overfitting.

For sensor placement optimization, GRASP was applied following the framework proposed by Resende and Silva [11]. Candidate solutions consisted of subsets of three chlorine sensors selected from the nine available sensor locations. In the construction phase, a semi-greedy algorithm generated candidate sensor placements. These solutions were subsequently improved through local search and path relinking. Candidate solutions were evaluated according to their contamination-detection performance, and high-quality solutions were stored in an elite pool. While the implementation follows the general GRASP framework, it was adapted to the sensor placement problem by restricting solutions to three sensors and by evaluating them according to contamination-detection performance rather than a generic optimization objective. The best solution identified by GRASP was then used for the final detector evaluation.

4 Evaluation

4.1 Experiments

All experiments were conducted on the **Net1** water distribution network from EPyT-Flow, shown in Figure 1. In all scenarios, arsenite (AsIII) contamination was injected at node 22 using an **EN_MASS**² source between simulation days 3 and 4. A constant chlorine source was placed at node 10, chlorine sensors were installed at nodes 10, 11, 12, 13, 21, 22, 23, 31, and 32, and each simulation was run for a total duration of 21 days. Two contamination strengths were considered: a weak contamination scenario with an arsenite injection rate of 100 and a strong contamination

²**EN_MASS** specifies a mass booster source in EPANET. A mass booster adds a fixed mass flow rate of a substance at a node, regardless of the node’s demand, and is commonly used to represent direct contaminant or disinfectant injection into the distribution system. Source strength is expressed as mass per unit time [6].

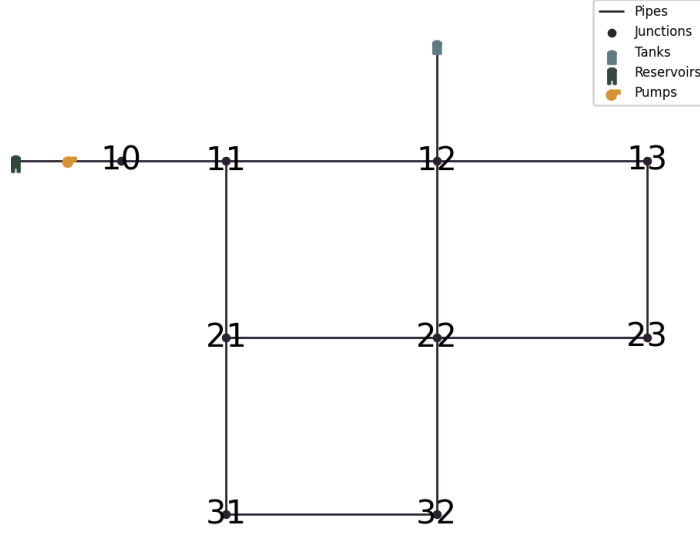


Figure 1: Network Topology of Net1

scenario with an injection rate of 100 000, corresponding to a 1000-fold increase in contamination intensity. Both the linear predictor and the ANN-based detector were evaluated under these conditions. Furthermore, GRASP was applied to each approach to investigate sensor placement optimization and assess whether reliable detection performance could be maintained with a reduced number of chlorine sensors.

4.1.1 Linear predictor

In the first experiment, we implemented the linear predictor exactly like it is explained in Sec. 3. However, the results of that experiment were not satisfactory, which is why we conducted further experiments.

The first problem was that the regression model always predicts a chlorine value that is different from the values before since it is technically unable to predict a constant value. However, the chlorine value stays constant sometimes in our injection scenario. For that reason, the algorithm might recognize anomalies, although the chlorine value did not change at all, which cannot happen if there was an arsenic contamination. Therefore, we implemented a correction of the predicted chlorine value to the actual chlorine value from the previous time step if the value stayed constant for the second experiment.

Unfortunately, the regression seems to predict the chlorine values after arsenic

contaminations too. Various attempts to solve this problem did not result in satisfactory results. We tried to use data with a very strong arsenic contamination, to amplify the differences by applying a power of 10 to the differences, and to use a gliding time frame of 10 consecutive values each, to track the chlorine value’s development over a longer period. The results barely improved, making these adjustments insignificant.

4.1.2 Artificial Neural Network

An ANN-based detector was evaluated on both weak and strong arsenic-contamination scenarios. The detector consists of two components. First, an ANN predicts the chlorine concentration at each monitored node from the current flow measurements, the previous 48 chlorine measurements (corresponding to one day), and cyclical time-of-day features. Prediction residuals and dynamic thresholds are retained as diagnostic information. Second, a supervised ANN classifier determines whether an event is occurring. It uses the current chlorine and flow measurements, chlorine changes over multiple time scales, and time-of-day features.

The data set contains 20 simulated scenarios for each contamination strength. Sixteen scenarios were used for model development and four scenarios were held out for the final evaluation. Within the development set, the training scenarios were split again into fitting and validation scenarios. This validation split was used for early stopping and to calibrate the alarm threshold; consequently, the held-out test scenarios were not used for model selection. The chlorine-prediction ANN used three hidden layers with 128, 64, and 32 neurons, while the event classifier used one hidden layer with 32 neurons.

In addition, GRASP was applied to the ANN detector to investigate sensor placement. At most three of the nine available chlorine sensors could be selected. For each candidate sensor subset, the ANN detector was trained using the same configuration and evaluated according to validation BA. GRASP combined randomized construction, local search, and path relinking, and evaluated 40 candidate subsets for each contamination-strength data set. After the best subset had been selected, its detector was retrained using all outer training scenarios and evaluated only once on the held-out test scenarios.

4.2 Results

A quantitative comparison between the ANN-based detector and the linear predictor is reported in Table 1. The table highlights the superior balanced accuracy, specificity, and chlorine-prediction performance achieved by the ANN across both contamination scenarios.

Method	BA	Recall	Specificity	FPR	Precision	R_{Cl}^2
ANN, weak contamination	0.705	0.635	0.774	0.226	0.176	0.909
ANN, strong contamination	0.545	0.538	0.551	0.449	0.196	0.818
Linear predictor, strong contamination						
ANN + GRASP (3 sensors), weak contamination	0.667	0.705	0.629	0.371	0.126	0.936
ANN + GRASP (3 sensors), strong contamination	0.636	0.457	0.814	0.186	0.336	0.869
Linear predictor, strong contamination	0.545	0.538	0.551	0.449	0.196	0.818
Linear predictor + GRASP (3 sensors), strong contamination	0.495	0.476	0.514	0.486	0.197	0.847

Table 1: Comparison of ANN and linear-predictor performance. GRASP configurations use three selected chlorine sensors. The linear-predictor baseline uses its best tested configuration (strong contamination, power transformation, and a 10-residual sliding window).

4.2.1 Linear predictor

The linear predictor method barely differentiated between anomalies and non-anomalies for all the experiments. This is because the linear predictor always predicted values that are very close to the actual values, as Figure 2 shows.

The only parameters influencing the anomaly detection were the threshold and the selected detection method. A higher threshold resulted in more detected anomalies, whereas a lower threshold led to fewer detections. The performance of the different methods also varied considerably. In ascending order of detected anomalies the normal and strong data methods detected no anomalies, the strong data and power amplification method detected approximately 30% of the anomalies, and the gliding timeframe method detected slightly more than 50%.

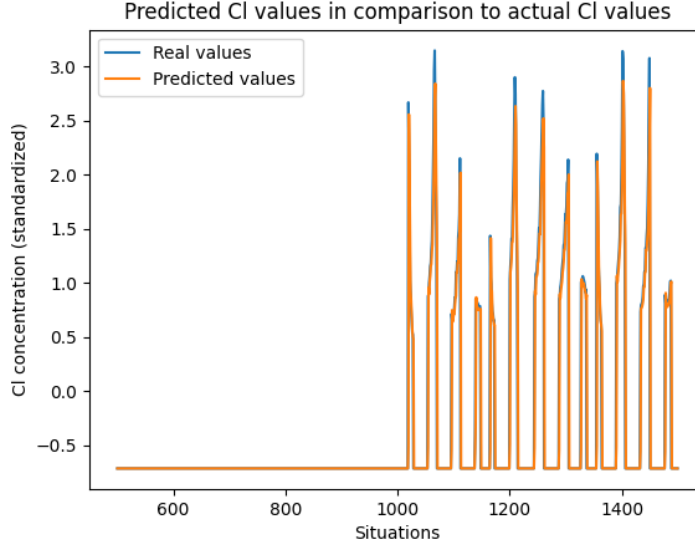


Figure 2: Accuracy of the linear predictor

However, there was no difference between actual contaminations and non-contaminations. So the strong data and power amplification recognized 30% of all situations as anomalies, regardless of whether there was an actual contamination. Therefore, the linear predictor works about as well as a baseline model.

A possible reason for this might be that the test data contains only a few arsenic contamination events, which may not be strong enough for the model to recognize them reliably. Furthermore, contamination and non-contamination events may not be linearly separable. Therefore, the regression model may overgeneralize by predicting values that are too close to the contamination values.

For sensor placement optimization, GRASP was applied as described in the methodology. The first experiment used the strong-contamination data set. However, because the linear predictor failed to detect any anomalies in this setting, all sensor placements achieved the same performance. Consequently, GRASP could not distinguish between candidate solutions and returned four equally good sensor placements instead of a single best solution.

To obtain a meaningful optimization problem, GRASP was subsequently evaluated using the strong-contamination data set together with the power amplification technique. In this configuration, the linear predictor was at least able to classify some situations as anomalies, allowing GRASP to differentiate between better and worse sensor placements. The algorithm selected nodes 12, 13, and 22 as the best

sensor configuration, achieving a true-positive rate of 9.57%. By comparison, using all chlorine sensors without GRASP resulted in a true-positive rate of only 5.61%. Thus, despite the overall limitations of the linear predictor, GRASP substantially improved the detection performance. Since GRASP includes randomized components, repeated runs may produce different sensor configurations with similar performance.

4.2.2 Artificial Neural Network

Unlike the linear predictor, the ANN detector was able to distinguish contamination events from normal operating conditions. For weak contamination, the ANN achieved a BA of 70.46%, a recall of 63.47%, and a specificity of 77.44%. Its ROC-AUC was 76.88%, indicating that the predicted event probabilities contained useful discriminative information. However, precision was only 17.59% and the false-alarm rate was 22.56%. Thus, although the detector identified many contamination events, it still generated a considerable number of false alarms. The average detection latency was 2250 seconds, corresponding to 37.5 minutes.

For strong contamination, the detector achieved a BA of 65.44%, a recall of 44.97%, and a specificity of 85.92%. The false-alarm rate was lower than for weak contamination, at 14.08%, and precision increased to 39.65%. The ROC-AUC was 72.82%. Nevertheless, recall was lower and the average detection latency increased to 5400 seconds, or 90 minutes. Therefore, a stronger contamination did not automatically lead to earlier or more frequent alarms in the simulated scenarios.

The chlorine-prediction component achieved a coefficient of determination of approximately 0.91 for both data sets, showing that chlorine concentrations could generally be predicted accurately. However, accurate chlorine prediction alone is not sufficient for event detection: the classifier must identify the relatively small changes associated with contamination without classifying normal operational variation as an event.

When comparing these results with those reported by Arad et al. [12], it should be noted that the two approaches were evaluated under substantially different conditions. Arad et al. employed six water-quality indicators, with a separate ANN model trained for each parameter, and subsequently combined their outputs using Bayesian alarm fusion. Their method was able to detect between 80% and 100% of simulated contamination events depending on the selected decision rule. By com-

parison, the recall obtained in this work ranged from 44.97% to 63.47%. Although the detection performance is lower, our approach relies only on chlorine and flow measurements and therefore uses significantly less information. This suggests that a reasonable level of contamination detection can still be achieved with a substantially reduced sensing infrastructure.

GRASP selected three sensors in both experiments. For weak contamination, the selected nodes were 11, 12, and 21. On the held-out test set, this configuration achieved a BA of 66.67%, a recall of 70.48%, and a specificity of 62.86%. For strong contamination, GRASP selected nodes 22, 23, and 31, obtaining a BA of 63.57%, a recall of 45.73%, and a specificity of 81.40%. These results show that the detector can still operate with only one third of the originally available chlorine sensors.

However, the GRASP sensor-selection experiment cannot be directly compared with the standard ANN. GRASP evaluates a restricted configuration (three sensors) and selects the subset based on validation performance, whereas the standard ANN uses the full sensor set (nine sensors). Therefore, differences in performance may be caused by both the number of sensors and the selection procedure based on performance, rather than by GRASP alone.

Finally, GRASP was also tested for ANN architecture search. In the synthetic demonstration, it reduced the number of classifier parameters from 9,985 to 2,559 while maintaining a comparable validation BA. This indicates that GRASP can be used not only for sensor placement but also to reduce the computational complexity of the ANN.

Table 1 summarizes the performance of the proposed ANN-based detector and compares it with the best-performing linear-predictor configuration. The comparison highlights the benefits of the ANN in terms of balanced accuracy, specificity, and chlorine-prediction quality.

5 Conclusion

This work investigated data-driven approaches for arsenic contamination detection in water distribution networks using chlorine measurements as indirect indicators of contamination events. Two detection methods were evaluated on the **Net1** network from EPyT-Flow: a linear predictor based on linear regression and an Artificial Neural Network (ANN). In addition, the Greedy Randomized Adaptive Search

Procedure (GRASP) was applied to optimize chlorine sensor placement.

The results show that the linear predictor is not suitable for reliable arsenic contamination detection in the considered scenario. Although several modifications were investigated, including stronger contamination scenarios, power amplification, and gliding time frames, the model was unable to effectively distinguish contamination events from normal operation. Consequently, the linear predictor mainly serves as a baseline for comparison.

In contrast, the ANN-based detector successfully identified contamination events and clearly outperformed the linear predictor. The best-performing ANN achieved a Balanced Accuracy (BA) of 70.46%, a specificity of 77.44%, and a detection latency of 37.5 minutes for weak contamination events. Furthermore, the chlorine-prediction component achieved a R^2 of approximately 0.91, indicating that chlorine concentrations could be modelled accurately. These results demonstrate that nonlinear machine-learning models are better suited to capture the complex relationships between hydraulic conditions, chlorine concentrations, and contamination events.

The sensor-placement experiments further showed that effective monitoring can be achieved with substantially fewer sensors than the full network deployment. GRASP identified sensor configurations containing only three chlorine sensors while maintaining competitive detection performance. This suggests that data-driven contamination monitoring can be combined with optimization techniques to reduce infrastructure requirements and associated costs.

Although the detection performance of the proposed framework (recall of 44.97%–63.47%) remains below that reported by Arad et al. [12] (80%–100% detection rates), representing a reduction of approximately 20 to 55 percentage points, our approach operates under considerably more restrictive sensing conditions. While Arad et al. combine six different water-quality indicators, the present framework relies only on a single water-quality indicator, chlorine concentration.

Furthermore, the integration of GRASP enables the simultaneous optimization of sensor placement, allowing effective monitoring with a substantially reduced sensing infrastructure. These findings demonstrate that meaningful arsenic contamination detection remains possible even when only limited monitoring information is available.

Future work should evaluate the proposed ANN-GRASP framework on larger and more realistic networks, investigate additional contamination scenarios and con-

tamination types, and explore advanced machine-learning architectures. Further improvements could also focus on reducing the false-alarm rate and on incorporating additional hydraulic and water-quality information to increase detection reliability.

In summary, the results indicate that the combination of ANN-based contamination detection and GRASP-based sensor optimization constitutes a promising approach for efficient and cost-effective water-quality monitoring in water distribution networks.

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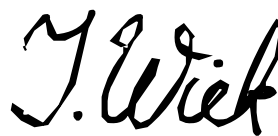
Affidavit

We hereby declare that we have written this report independently and that we have prepared the data sets, drawings, sketches and graphical representations provided independently. We have not used any sources other than those indicated and have always indicated those parts of the report as borrowed, which are taken from other works, including tables and illustrations used, while stating the source. This report represents the personal writing of and preparation by ourselves, and was not generated by any large-scale language model, such as ChatGPT or similar. This work, to the best of our knowledge, has not been submitted to any examination authority in the same or a similar form.

Bielefeld, 11.08.2026



Esther Giuliano



Julian Wick

Contributions

The contributions of the individual authors were as follows:

- Literature review and chemical background research: Both authors
- Data generation and preprocessing: Julian Wick
- Linear predictor implementation, experimentation, evaluation and GRASP implementation: Julian Wick
- ANN implementation, training, experimentation, evaluation and GRASP implementation: Esther Giuliano
- Analysis and discussion of results: Both authors
- Writing and revision of the report: Both authors